

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-a4e50636d8292965b.jsonl`  
- SHA-256: `de2097cbd55498aa113bab8acea9506b51440125687f24dd6d5208edfb435f74`  
- Source modified: `2026-06-21T13:58:34+00:00`  
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- project: `wf_b6ef6275-de5`  
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T13:56:13.271Z)

Read the OWNED-MODEL + STRATEGY track docs in the engine repo (C:/Users/avidu/Projects/badminton-highlight-indexer) and extract status + next steps. Read docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (the authoritative roadmap + its Phase-0 greenlit milestone + tracker), docs/COMMERCIALIZATION.md, docs/PMF_MARKET_RESEARCH.md + docs/PMF_FIELD_SIGNALS.md, docs/DEFERRED_HARDENING_PLAN.md (esp. #16 async jobs ? note it's DONE now ? and other open items), docs/ROADMAP.md, docs/BACKLOG.md. Capture next steps + which are owner-gated (Phase-0 greenlight, training-data opt-in, licensing). Return structured JSON (track="owned-model+strategy").

2. assistant (2026-06-21T13:56:16.162Z)

I'll start by pulling the repo for freshness, then read the owned-model and strategy track docs.

3. user (2026-06-21T13:56:19.029Z)

Already up to date.

4. assistant (2026-06-21T13:56:21.220Z)

Now let me read the docs in parallel.

5. user (2026-06-21T13:56:22.191Z)

```
1      # Owned-model implementation plan (roadmap) ? DRAFT for owner review  
2  
3      **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner  
4      greenlights a specific  
5      milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a  
6      training source).  
7      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +  
8      learned prototype  
9      (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy  
10     workflow (2026-06-09).  
11  
12     ## Design principle (owner directive): *envision now, build later*  
13     Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be  
14     **architecturally envisioned  
15     now** so we never hit a mammoth refactor. We **design the seams** (data contracts,  
16     export boundaries, the  
17     event store) up front and **implement behind them** when each milestone is greenlit.  
18     Punting a *feature* is fine;  
19     punting its *seam* is not.  
20
```

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14     ## North Star (reframed per PR #61 review)
15     - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16     recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17     - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18     contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19     - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20     roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21     - **The moat = the consented dataset + the eval harness, not the weights.**
22
23     ## Hard guardrails (non-negotiable ? "never corrupted")
24     - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25     (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26     **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27     - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28     were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29     - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30     - ? **Base-model pretraining provenance is unauditable for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31     training" rule is enforceable at *our fine-tuning data* layer, not the base's
pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(runs SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but
its Prohibited-Use posture is
43     unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not
the primary extractor.
44     **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is
cheap *only within
45     timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-
specific).
46     - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss
strict temporal boundaries**
47     (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse
regions, not accurate boundaries").
48     **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-
risked on our corpus (held-out
49     AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params
(trivially compressible later),

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50         and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So:
**head now ? VLM later**,
51         strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes,
head refines boundaries) over
52         wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur
footage may differ ? re-measure.)*
53
54         ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55         **Punt the conversational *feature*** until extraction quality is solid (current LOVO
~0.583 ? "very high"; a chat UI
56         over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn**
(expensive; bad at exact
57         counting; the strong teachers are license-forbidden). **But build the *seam* now** (the
brief requires it):
58         - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
59         spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-
SQL + ffmpeg clip-cut). The
60         cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a
61         VLM job** (SQL counts exactly where VLMs hallucinate).
62         - **Designed-now seam:** make the **per-video event-index schema a first-class data
contract** alongside the M0.1
63         corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count,
ending_reason/fault_tags,
64         derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy
(e.g. `service_fault`) up front**
65         even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration):
`play_segments` today has
66         `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights`
isn't a table yet ? so it's an
67         *extension with new columns + a highlights table*, not a no-op. **Punt** the NL
parser/agent until quality clears the bar.
68         - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our
structured index (allowed ?
69         it reasons over our data, it is not trained on its outputs).
70
71         ## Multisport (single sport-agnostic model)
72         - **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned
across sports (sport = an input
73         feature / per-sport calibration). The collector + indexer are already sport-generic. ?
**ACTION: rename the repo off
74         "badminton" ASAP** (owner emphatic) ? consistent with this.
75         - **What multisport changes = DATA + DETECTORS (scale up, the accepted "good
problem"): ** each new sport needs its own
76         labeled matches (split by match) + a **clean trajectory detector** to feed the head's
features. **The binding blocker
77         is per-sport detectors: ** WASB (badminton) has the unresolved **C3** checkpoint-
provenance issue, and **clean MIT/
78         Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet
identified. ** Sequence by public-data
79         richness: **badminton ? tennis/TT ? pickleball/padel.**
80         - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81         and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82         concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport
83         corpus exists. Don't cite RacketVision as evidence.
84
85         ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
shell
86         - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
offline/batch, no latency hot path.
87         - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-

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process (ffmpeg, WASB-in-WSL, cloud
88     GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
single-digit-ms wins dwarfed by
89     ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
harness + provider registry.
90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91     Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92     runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93     target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95     separate service. **None exists today.**
96
97     ---
98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
*greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg . TAL
ActivityNet-JSON . VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
(system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
enforced; **no Gemini-sourced rows in any training view**.
108
109     ### M0.2 ? Grow the corpus . **DONE (n=6), running** . *owner action + my scoring loop*
110     6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-
match + LOVO recorded; the
111     "contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club
match ? margin. **Bottleneck =
112     data + calibration, not method.** Keep growing per new sport.
113
114     ### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) .
*greenlight item*
115     - **(i) PURGE the Gemini-silver TRAINING path** ? the live no-paid-LLM-training
violation is
116     **`backend/eval/distill_local.py`** (it trains a classifier on Gemini-silver CSV
labels ? `output/local_rally_model.json`)
117     and the threshold distillation in `calibrate_local.py`. (NB:
`backend/eval/training.py::label_candidates` is a *pure,
118     source-agnostic* function ? there is no store by that name; the earlier
"training.label_candidates" pointer was wrong.)
119     Nothing Gemini-trained is in git (artifacts live under gitignored `output/`), but the
path that regenerates them is one
120     command away ? retarget to human + open-teacher (own-model/WASB) labels; Gemini =
eval/reference/fallback only.
121     - **(ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint
provenance is a **commercial-ship
122     blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a
clean MIT/Apache detector swap.
123     ? **own-weights is an un-scoped NEW workstream, not ROADMAP Phase 4:** per ADR-002 our
temporal labels can't train the
124     shuttle detector (it needs per-frame coordinate labels we don't have). **Multiplies
per sport** (incl. any TT-on-WASB use).

```

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125     Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
126
127     ### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) . *greenlight item*
128     Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
**ActionFormer (MIT, 1:1
129     rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not*
the Commons-Clause repo),
130     reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item):
a FLOOR of "beat the honest n=6
131     heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU**
target for "very high P/R" (e.g.
132     F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory.
Trains on the GPU/WSL box.
133
134     ---
135
136     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
137     Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it
becomes the rally/highlight
138     extractor (local, MIT). If not, the gap (more data/features) is the next signal ? no
overfitting to declare victory.
139
140     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus +
explicit go)
141     - **Gen-1:** concatenate fully-local cue channels onto the head. ? **Audio is NOT a
promising cue here** ? our E1 probe
142     (PR #35) found it weak (~2.2x discrimination, AUC ~0.6) and the foreground-audio idea
also failed this session; keep it
143     parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet
first).
144     - **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a
temporal-IoU-vs-golden reward,
145     our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on
Modal (no-access) / RunPod-Secure.
146     - **Hard pre-serving gates (unmeasured, load-bearing):** the **$/call crossover** (self-
hosted VLM vs Gemini) *** a real
147     VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples
use 6?8 GPUs ? **budget is the
148     largest open risk and is currently unquantified**). Don't stand up an owned serving
stack until these clear.
149
150     ---
151
152     ## Decisions / status
153     | # | item | status |
154     |---|---|---|
155     | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
156     | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
157     | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
158     | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
159     | 5 | Rename repo off "badminton" | **owner action** ? |
160     | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
161
162     ## Open risks / unknowns (none block Phase-0 research)
163     Budget/cost to reach Gen-2 **unquantified** (largest risk) . ship-gate "very high P/R"
target undefined . per-sport clean
164     trajectory detectors not yet identified . base-model pretraining provenance unauditale
(risk-acceptance) . real VLM
165     VRAM/token cost . single-model-multisport-temporal generalization unproven . DPA + no-
train (minors-excluded) for cloud
166     training . on-device 4B latency on 8 GB unmeasured.
167
168     ---

```

```
169      *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing
discipline holds: the committed
170      next platform slice remains the async job model (single-tenant); only the M0.1?M0.4
foundation is safe to start.*
171
```

6. user (2026-06-21T13:56:22.668Z)

```
1      # Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETED)
2
3      **STATUS: COMPLETED** (2026-06-14). This planning doc's three items are all dispatched:
4
5      - **#16 Async job model ? DONE** (PR #87; hardened by the self-scheduling drain loop in
#149).
6      - **#7 Packaging modernization ? DONE** (PR #167: PEP 621 `pyproject.toml`, `setup.py` ?
`scripts/doctor.py`, `indexer` console entry point, CI switched to `pip install -e .[dev]`).
7      - **#13 Train/eval separation guardrails ? moved to [BACKLOG.md](BACKLOG.md)** (?; the
concrete `splits.py` / `assert_disjoint` match-level-split design is preserved there).
8
9      Full archived snapshot (the original plan + per-item design/risks/verification):
[archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md).
10
11      _(This top-level file is a pointer stub kept so existing links to
`docs/DEFERRED_HARDENING_PLAN.md` still resolve; the content now lives in the archive.)_
12
```

7. assistant (2026-06-21T13:56:27.400Z)

DEFERRED_HARDENING_PLAN is now a completed pointer stub (#16 async jobs DONE, #7 packaging DONE, #13 moved to BACKLOG). Now let me read the strategy docs.

8. user (2026-06-21T13:56:28.314Z)

```
1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
counsel . ? future
```

30 Priority: ****P0**** (do first) . ****P1**** . ****P2**** (lowest). Rows are grouped ****Pending ? In-flight ? Completed****, each sorted by priority. (C# IDs are stable identifiers ? they don't imply order.)

31

32 | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |

33 |---|---|---|---|---|---|

34 | | ****? PENDING (needs action / not started)**** | | | |

35 | C5 | ****Output codec royalties**** | ****P0**** | ? | Highlights are still encoded ****H.264 + AAC**** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure remains. | Switch output to ****AV1 + Opus**** (royalty-free) on an ****Ada-class GPU (L4/L40)**** for HW encode. See §4. Decision: provision Ada-class GPU? |

36 | C9 | ****Packaging / distribution**** | ****P0**** | ? | `pyproject.toml` (PEP 621, hatchling) added; the diagnostic `setup.py` renamed to `scripts/doctor.py`; `indexer` console entry point; CI installs `pip install -e .[dev]`. torch/torchvision stay an out-of-band `--index-url` install (PEP 621 can't pin CUDA wheels). | Implemented per `DEFERRED\_HARDENING\_PLAN.md` (#7). Remaining: publish `obsreport` so the `reporting` extra is pip-installable. |

37 | C12 | ****SaaS foundations (async/auth/tenancy)**** | ****P1**** | ? | Async job model shipped (`DEFERRED\_HARDENING\_PLAN.md` #16, PR #87); auth/tenancy still future. | Multi-user/platform strategy in `PLATFORM\_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration). |

38 | | ****? IN-FLIGHT (in progress / watch)**** | | | |

39 | C4 | ****Runtime AI strategy (A/B/C + owned model)**** | ****P0**** | ? | Live = ****C**** Gemini 2.5 Flash, paid tier. Long-term = ****own a self-hosted generative video-QA model**** (fine-tuned open VLM) as primary, ****paid Gemini always-ready as fallback**** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model ****after**** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |

40 | C13 | ****Beachhead validation**** | ****P0**** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends ****India badminton academies**** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |

41 | C8 | ****Golden-set vendoring spend**** | ****P1**** | ? | ADR-006: ****Roora (Bengaluru)**** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |

42 | C3 | ****Shuttle-tracking weights provenance**** | ****P1**** | ? ? | WASB-SBDT ****code is MIT (clear)****; distributed ****checkpoints are academic-data-trained**** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial). | ****Ship-gating**** blocker (multiplies per sport). Resolve via an own-trained detector ****or**** a clean MIT/Apache detector swap ? NOTE: ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |

43 | C14 | ****Owned-model base licensing + training-data policy**** | ****P1**** | ? ? | Owned video-QA model must build on a ****commercial-clean (MIT/Apache-2.0/BSD), video-capable**** base ? primary ****Qwen3-VL (Apache-2.0)****, fallback ****InternVL3 (MIT)****. ****Gemma 4 = AMBER / bench-only**** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build target yet). ****Reject Gemma 3**** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). ****Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (competing-model clause). The moat is the ****consented dataset + eval harness****, not the model. | ****MIT/Apache-2.0/BSD**** for code+data+weights; ****open-model teachers only****; separate explicit ****training opt-in****; ****exclude minors****; per-clip consent ledger. See `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (authoritative). |

44 | C6 | ****Inbound HEVC decode**** | ****P2**** | ? ? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is ****cheap**** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48 | C2 | ****`supervision` ByteTrack pin**** | ****P2**** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

```

49      | | **? COMPLETED** | | | |
50      | C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8
replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL
code/weights in the stack. | Done. Keep new deps permissive. |
51
52      ---
53
54      ## 2. Licensing & dependency posture (refreshed risk table)
55
56      Verdicts updated against the current stack (`requirements.txt` + code). Original
57      2026-05-30 verdicts and the 25-claim verification transcript are in
58      `archives/MONETIZATION_AUDIT.md`.
59
60      | Dependency | License | SaaS obligation? | Current verdict | Notes |
61      |---|---|---|---|---|
62      | ~~`ultralytics` + `yolov8n.pt`~~ | AGPL-3.0 | Yes (§13) | ? **REMOVED** | Replaced per
ADR-005 ? no longer in the stack. |
63      | **torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen
replacement; person = COCO class (see `backend/pipeline/detectors/person_detector.py`
`_PERSON_CLASS_ID`). |
64      | **`supervision`** (ByteTrack) | MIT | No | ? LIVE | Pinned `<0.30.0` ? migrate before
bump (C2). |
65      | **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open
(C3). |
66      | **MonoTrack / YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
67      | **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH |
Must bill the paid tier to keep user video private (C4). |
68      | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? |
Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |
69      | ffmpeg/ffprobe, google-genai, torch, opencv, numpy,
fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as
subprocess / imported; all permissive. |
70      | **`obsreport`** (reports) | permissive (private) | No | ? soft dep | Pin the
`git+ssh@vX` tag once the repo is published (BACKLOG). |
71
72      **Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)
73      are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
74      and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
75
76      ---
77
78      ## 3. Runtime AI strategy ? A/B/C
79
80      The semantic rally-boundary step is the one external/metered dependency. Three paths
81      (from the audit), with current standing:
82
83      - **(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
84      classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85      cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86      shuttle-motion, decided in ADR-004). **Most engineering.** ? **target**
87      - **(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88      per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89      - **(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
90      ongoing metered cost + external dependency. **? current live path.**
91
92      Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93      inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94      OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96      ---
97
98      ## 4. Codec / output economics (the live gap ? C5)
99
100     **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101     codecs it implements. Royalties attach to the act of encode/decode.

```



```

102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](archives/field-pmf/PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:
142 1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143 players, formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
144 who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
145 not the individual; needs INR-tuned pricing **and a Kannada/Hindi product** ? see
146 [`I18N_PLAN.md`](I18N_PLAN.md) (Bangalore-first localization, by design).
147 2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8%
148 YoY) but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
149 head-to-head, not greenfield.
150 3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China
151 needs local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
152
153 Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B
154 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/
155 Spiideo; comparable to SwingVision software-only).
156
157 ### PMF risks (from the readiness review)
158 - **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
159 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
160 narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
161 - **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;
162 per-rally tags, stats, and a correction loop are what make it sticky (C10).

```

```

162 - **Golden labels are the recurring blocker** across the offline segmenter, audio
163 fusion, and measurable quality ? the highest-leverage investment (C8).
164
165 ### Refined priority order (quality/data phase ? "focus on quality once new videos
land")
166 Carried from the readiness review; commercialization items above are tracked in
parallel.
167
168 1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos):
169 ingest the new batch, expand calibration/CV/regression baselines, make label
170 contribution + safe retraining easy and non-contaminating (eval/training separation).
171 2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract
172 shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics
land.
173 3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/
174 min-duration + `rally_gate`; validate WASB on real video; trial audio fusion
(ADR-007).
175 4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ?
176 `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
177 5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ?
178 validators/report rules; "your video is good enough" UX.
179 6. **Multi-sport maturation + recording best practices.**
180 7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
181 8. **Polish & distribution** (C9): installer story, first-run path, codec/output,
economics.
182
183 **Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
184 (flywheel + quality iteration on real data), then unification (#2) and deeper value
185 (#4).
186 ---
187
188 ## 7. Open questions needing owner / counsel sign-off ??
189
190 1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
191 insufficient (fallback). Quote-only.
192 2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
193 confirm each (C3).
194 3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
195 figures with counsel before relying on the "small SaaS under threshold" reading
(C5/C6).
196 4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
197 throughput (C7).
198 5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
199
200 ---
201
202 ## Changelog
203
204 - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
205 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
206 "beachhead validation" workstream. Wired the summary into §6.
207 - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
208 audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
209 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
210 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
211 `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
212 exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
213
214 ---
215
216 ## See also
217 - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
beachhead; cited).

```

218 - `archives/MONETIZATION\_AUDIT.md` ? licensing/economics research-of-record (sources + verification transcript).
219 - `archives/COMMERCIAL\_READINESS\_REVIEW.md` ? superseded assessment (history).
220 - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor process).
221 - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
222 - `DEFERRED\_HARDENING\_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
223 - `GOLDEN\_SET\_VENDORING.md` ? vendor process + Roora pilot (C8).
224 - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
225

9. user (2026-06-21T13:56:28.803Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

10. assistant (2026-06-21T13:56:36.751Z)

PMF_MARKET_RESEARCH.md isn't at docs top level ? the COMMERCIALIZATION doc references it at `archives/field-pmf/PMF\_MARKET\_RESEARCH.md`. Let me locate the PMF docs and read the remaining ones.

11. user (2026-06-21T13:56:37.087Z)

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12. user (2026-06-21T13:56:37.370Z)

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13. user (2026-06-21T13:56:37.424Z)

docs\BACKLOG.md

14. assistant (2026-06-21T13:56:40.171Z)

The PMF docs live in `docs/archives/field-pmf/`. Let me read those plus ROADMAP and BACKLOG.

15. user (2026-06-21T13:56:41.125Z)

1 # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3 > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;

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4 > validate live before relying on them for investment/pricing decisions. This is the
5 > market-research-of-record behind the PMF section of the living
6 > [COMMERCIALIZATION.md](COMMERCIALIZATION.md) tracker ($6). Product framing:
7 > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8 > self-recorded, static-camera, full-court amateur racket-sports video and
9 > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10 > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12 > Confidence convention. Figures from named industry reports, federations, or
13 > company pricing pages are tagged with source + year. Inferred items are tagged
14 > (estimate) / (low confidence). Emerging-category market sizes vary widely
15 > between vendors; ranges are shown rather than false precision.
16
17 ---
18
19 ## 1. Executive summary
20
21 The product sits at the intersection of three favorable trends: (a) a fast-growing
22 sports analytics market (~USD 5.6B in 2025, ~18-22% CAGR to USD 24-36B by ~2034
23 depending on vendor); (b) a structural shift to amateur self-recording on action
24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25 segment); and (c) strong racket-sport participation ? 220M+ badminton players,
26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27 2024, +45.8% YoY).
28
29 The central tension is the static-camera constraint: the tool works best on fixed,
30 full-court amateur footage and fails on broadcast/panning footage. This is both a
31 moat (most competitors target team sports / broadcast) and a filter on the addressable
32 user ? the customer must already set a camera on a tripod behind the court. That
33 behavior is concentrated in coaches/academies, clubs, and serious enthusiasts, not
34 the median casual player.
35
36 Recommended beachhead (strongest single): badminton coaching academies and
37 serious club/competitive players in India ? a large, rapidly formalizing academy
38 ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39 willingness to pay for coaching, an underserved tooling gap (incumbents are
40 team-sports/broadcast), and a direct-but-immature competitor field. The product is
41 native badminton-first ? a genuine edge here. Secondary: (2) US/global pickleball
42 players & clubs (capital-rich, fastest-growing, but SwingVision already strong);
43 (3) badminton academies/clubs in East/SE Asia + Denmark (highest density).
44
45 Pricing clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46 and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47 software-only BYO-camera model undercuts the hardware players; realistic anchors
48 (estimate): B2C USD 60-180/yr; B2B academy/club USD 300-1,500/yr per location.
49
50 Top risks: (1) amateurs won't reliably set up a tripod ? TAM collapses to
51 coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52 variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53 China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55 ---
56
57 ## 2. Market sizing & trends
58
59 ### 2.1 Sports analytics / sports-video market
60 - 2025 size: ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61 Insights); other vendors USD 5.6-6.0B. (2025)*
62 - Growth: CAGR 15.7-22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). (reports 2024-2025)*
65 - Video analytics is the leading segment within sports analytics (Grand View /
66 Fortune BI) ? the directly relevant sub-market.
67 - Interpretation (estimate): the amateur/club auto-highlight slice is a small but
68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The

```

product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- **2024 size:** ~USD 6.5B (GMInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
- **Sports is the largest end-use segment** of action cameras (GMInsights, 2024).
- **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:** the capture-side behavior the product depends on is already widespread and growing ? *for the segments that record* ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)	
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	Badminton ? China ~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton	India 20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	Badminton ? Indonesia National sport; ~1M club players industry (est.)	
Badminton	Denmark 100,000+ registered club members (Europe #1) industry (2023)	Badminton ? Japan/Korea Major school-club + workplace sport industry (2023)	
Tennis	**106M players (2024)**; +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	Tennis ? Asia ~1/3 of players (33M of 87M) ITF (2021)	
Pickleball	**US** **19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	Pickleball ? US courts ~70,000 courts, +55% places to play YoY SFIA / Pickleheads (2024)	
Pickleball	global IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim *(low confidence)* various (2024?2025)	Table tennis	30M+ players; 227 member associations ITTF (current)
Cricket	2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders) Cricket Rise / Nielsen (2024?2025)		

- **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR (Market.us/Custom Market Insights, 2024?2025).
- **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

Geographic read: badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as *fandom* but its amateur self-recording + static-full-court fit is weak (large field, not a fixed-frame court). For *this product's static-camera fit*, badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.

3. Customer segment evaluation

Segment	Market-size signal	Willingness to pay	Accessibility	Static-camera fit
B2C/B2B	Verdict			
Individual amateur/enthusiast	Largest raw TAM (100s of millions) **Low** ? competes with free; ~USD 5?15/mo ceiling High via app stores; high CAC/churn **Mixed** ? most won't set a tripod B2C Volume play, weak economics; use as funnel			
Coaches & coaching academies	India academy boom; thousands across Asia **High** ? already sell coaching; USD 300?1,500/yr plausible Reachable via federations/distributors/social **Strong** ? already film fixed-angle B2B/B2B2C **Best fit** ? buyer, recorder, payer aligned			
Clubs & leagues	Denmark 100k+; thousands of clubs Asia/EU Medium-high; USD 500?3,000/yr Medium ? committee sales cycle **Strong** B2B/B2B2C Strong secondary; longer cycle			

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |
119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |
120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |
121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor ? panning = failure mode**** | B2B | ****Avoid**** ? violates core constraint |
122
123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.
127
128 ---
129
130 **## 4. Competitive & pricing benchmarks**
131
132 | Product | Category | Model | Price | Notes (year) |
133 |---|---|---|---|---|
134 | ****Veo (Cam 3)**** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |
135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300
(Pro) | Camera leased free with sub (2025) |
136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD
69?167/mo | Institutional/league; 10,000+ clubs (2025) |
137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ?
6,100/yr (club) | Institutional budgets (2025) |
138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports
incumbent (2026) |
139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD
179.99/yr****; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |
140 | ****BadPro+**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "
one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights
(2025) |
141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |
142
143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses ***pay-per-use*** ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.
152
153 ---
154
155 **## 5. Recommended beachhead**
156
157 Ranked by ****WTP × participation density × existing recording behavior × static-camera
158 fit × competitive whitespace.****
159
160 **### #1 (strongest) ? Badminton coaching academies & serious players, **India****
161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is ***native badminton-first***.
165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not
166 the individual.
167
168 **### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)****
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;

strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:****
****SwingVision** already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, ****East/SE Asia** (China, Indonesia, Malaysia, Japan, Korea) + Denmark**

Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording.
****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer alignment), use ****enthusiast B2C** as the funnel**, validate a ****club/academy B2B2C motion****, then expand to ****pickleball-US**** (out-build SwingVision) and ****badminton-Asia/Denmark**** (density).

6. Risks & open questions (what would invalidate the thesis)

- **Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a fixed camera, B2C TAM collapses to coaches/clubs. ***Validate:*** observe 3?5 academies + 50?100 enthusiasts on whether they already record fixed-angle.
- **Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap B2C WTP. ***Validate:*** fake-door pricing at USD 60?180/yr.
- **Capture-quality variance = the failure mode**** (gym lighting, off-center angle, occlusion, mixed resolutions). ***Validate:*** benchmark rally-detection P/R on 100+ real amateur clips; define a "good-capture" spec + auto-reject (ties to `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).
- **Monetization mismatch in best-participation markets**** (India/Indonesia/China are price-sensitive, hard to bill). ***Validate:*** local pricing/payments pilot; per-academy INR licensing.
- **Competitive response**** (SwingVision or a team-sports incumbent adds racket sports). ***Validate:*** ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
- **B2B sales-cycle drag**** for clubs/federations/schools. ***Validate:*** single-coach/academy self-serve before institutional deals.
- **Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

****Cheapest next steps:**** (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

7. Sources

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- Action camera market: <https://www.gminsights.com/industry-analysis/action-camera-market> (2024);
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 234 - Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
 235 <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
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 237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
 238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
 239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
 241 - Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing>
 ;
 242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
 243 <https://www.badproplusapp.com/>
 244

16. user (2026-06-21T13:56:41.655Z)

```

1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [ `PRODUCT_FIELD_ASSOCIATE_FLYER.md` ](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
7      (the recruiting flyer candidates see) .
8      [ `FIELD_VISIT_PLAYBOOK.md` ](FIELD_VISIT_PLAYBOOK.md)
9      (the hired associate's visit playbook: script + ground rules + Phase-2) .
10     [ `FIELD_VISIT_ANSWER_SHEET.md` ](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
11     [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
12
13     **Why this doc exists:** the playbook scripts the questions and the answer sheet
14     collects what was said; this doc fixes ? **before the first visit** ? what those
15     answers *mean* and what totals trigger which decision.
16     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
17     15 ambiguous conversations as whatever we hoped to hear.
18
19     ---
20
21     ## 1. The hypotheses under test (each maps to script questions)
22
23     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
24     |---|---|---|---|
25     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
26     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
27     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
28     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+ ? the
India set is field-sourced from the 2026-06-10 competitive audit; PMF §4's web table predates
it) leave a gap we fit | "we use X, it's fine, fully solved" | Q4 |
29     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
30     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
31     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |

```


32
33 Unit-economics floor behind H5: serving COGS on the Gemini path is ~\$0.30?0.60 per
34 hour of video (? 25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
35 roughly 75?150 to process, so **?300/match has real margin, ?100/match is
36 break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
37
38 ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
39
40 **Scoring is done by the owner (or a Claude session) at transcription ? the
41 associate only captures answers** (the sheet's score grid is marked OFFICE USE).
42 Score each hypothesis **from what they said and showed, not what you inferred**.
43 When between two grades, take the lower one.
44
45 | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
46 |---|---|---|---|
47 | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
48 | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
49 | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
50 | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it,
satisfied, sticky |
51 | **H5 WTP** | names ??300/match **BEFORE the ladder was offered** (the sheet's tick-box
records this), or proposes a budget framing of their own whose per-match equivalent is ??300
(see counting rule below) | names a number only **AFTER** the ladder (anchored ? even
?300/?500), or volunteers ~?100 | refuses to name a number / "should be free" / volunteers <?100
|
52 | **H6 multi-court** | complains about it in their own words before/without prompting |
agrees when asked | multi-court footage already works for them |
53 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |
54
55 **The H5 counting rule ("H5 at ?300+"):** a visit counts toward the "?300+" tally
56 only when the ? number (a) is ?300 **and** (b) came *before* the ladder was offered
57 ? an anchored ?300 is the modal polite answer to a read-aloud 100/300/500 ladder
58 and proves nothing, so it scores AMBER and does **not** count. Budget-framed
59 answers convert at transcription: their monthly ? ÷ the matches/month they cited
60 (if they cited none, ÷4); the computed equivalent counts if ??300 and the
61 conversion is written into the rollup row. Stated WTP ? even clean ? remains
62 *talk*: the confirming step before any PROCEED-triggered build is the fake-door /
63 paid-pilot test already pre-agreed in PMF \$6 risk 2 and the C13 row.
64
65 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral*
66 signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2
67 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a
68 polite conversation, not a customer.
69
70 **Disconfirming-evidence quota:** every visit sheet must record at least one
71 verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a
72 sheet has only positives, the visit gets re-done or its evidence discounted in the
73 rollup (**never the associate's pay** ? they're paid per completed honest visit);
74 per the playbook's ground rule 4, a reasoned "no" is worth more than five vague yeses.
75
76 **Sampling note (read the verdict with this in mind):** the playbook sends the
77 associate to 10?15 *good* academies ? the verdict therefore reads "does the
78 beachhead work where it should work best?", not "what fraction of the market is
79 addressable". A PROCEED here licenses the next validation step, not a TAM claim;
80 the rollup's area/courts columns record what was actually sampled.
81
82 ## 3. Pre-registered decision rules
83
84 **Stopping rule (fixes the optional-stopping loophole):** the campaign targets
85 **n = 15** completed owner/head-coach sheets and may end earlier **only for
86 logistics** (associate availability, academies exhausted) ? **never because of how

87 the tallies look**. The 5-sheet check-ins below are progress reports, not verdict
88 opportunities; **the verdict is computed exactly once, at the final n**, and the
89 rollup records what that n was and why the campaign stopped. The count thresholds
90 below are written for n = 15 and apply unchanged at any final n in 10?14 ? which
91 deliberately makes the ?-bars *relatively harder* at smaller n (biases away from a
92 premature PROCEED) and the <-bars relatively harder to trip at n = 15 (biases away
93 from a premature FALSIFIED). Both biases err toward "keep investigating," the safe
94 failure mode. Counts, not percentages ? at this sample size we're reading
95 direction, not estimating proportions. "GREEN" below means GREEN on that
96 hypothesis row.
97

98	Verdict	Trigger (pre-registered)	What we then do
99	--- --- ---		
100	**PROCEED ? beachhead validated**	?6 H2-GREEN **and** ?4 H5 at ?300+ (per the S2 counting rule ? anchored numbers don't count) **and** ?3 H7-GREEN	Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); the **fake-door/paid-pilot WTP test** (PMF S6 risk 2) runs before money is spent on build
101	**PIVOT A ? "index the backlog" ? "record + index" service**	<3 H2-GREEN **and** ?6 visits with H3 GREEN-or-AMBER **and** ?3 H5 at ?300+	The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo
102	**PIVOT B ? partner/API motion**	H4 RED at ?5 academies (incumbents entrenched and liked)	Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship ¹
103	**PIVOT C ? payer is the parent, not the academy**	?4 visits where H5 is RED **and** the Q3 answer names parents/players as the watcher	Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer
104	**FALSIFIED ? beachhead assumption fails**	<3 academies record at all (H1) **and** 0 H7-GREEN	The "academies already record fixed-angle" premise (PMF S5 beachhead #1; the academy arm of S6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further

105

106 ¹ The template (surfaced by the 2026-06-10 competitive audit, not yet in PMF S4's
107 web-sourced table): **PB Vision** sells pickleball video analysis B2C at ~\$99/yr
108 *and* as an ~\$8/hr-of-video partner API through facility platforms like
109 **PodPlay** ? the engine is ours, the academy relationship belongs to whoever
110 already owns it.
111

112 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide
113 explicitly with the owner ? but **the tallies above get computed first**, so the
114 discussion starts from the pre-registered read, not from anecdotes.
115

116 **Competitor-intel side-channel (always-on):** any visit where a competitor is
117 named ? fill the answer sheet's competitor block (what it does, what it misses,
118 what they pay). ?3 consistent reports about one product = update
119 `PMF\_MARKET\_RESEARCH.md` S4 with field-verified rows (today's table is web-sourced).
120

121 ## 4. Mechanics
122

- 123 - One [`FIELD\_VISIT\_ANSWER\_SHEET.md`] (FIELD_VISIT_ANSWER_SHEET.md) per academy,
124 same day as the visit, in the associate's words + verbatim quotes.
- 125 - The owner (or Claude session) transcribes each sheet into one row of the rollup
126 table below (keep it in this doc or a sheet next to it); tallies refresh per 5
127 visits as **progress checks only** ? the verdict is computed once, at the final
128 n, per the S3 stopping rule.
- 129 - Demo asset: the associate carries the real compiled reel
130 (`output/MahadevpuraSingles\_highlights.mp4` , trimmed to ~60s) saved offline on
131 their phone, shown **only at the scripted moment (after Q6, before Q7)** ? *show,
132 never promise*. Demo reactions and improvement asks are **product feedback**, not
133 scored signals: they feed the build backlog verbatim, while the H1?H7 verdict
134 machinery stays anchored to the pre-demo answers (that's why the demo comes last;
135 the sheet records whether it was shown). Showing it right before Q7 also makes
136 the Phase-2 ask a reaction to something seen, not imagined ? a stronger
137 behavioral read.

```

138 - Student answers (sheet's student block) likewise feed product feedback and the
139 H3 (audience) gut-read; they don't enter the verdict tallies, and no student
140 contact details are ever collected.
141
142 ### Rollup table (one row per completed visit)
143
144 | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
145 |---|---|---|---|---|---|---|---|---|---|---|---|---|
146 | 1 | | | | | | | | | | | | |
147
148 ### Tally ? decision read-out (progress check per 5 visits; verdict ONCE at final n)
149
150 | Metric | Count | Threshold in play |
151 |---|---|---|
152 | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
153 | H5 at ?300+ (pre-ladder / converted only ? §2 counting rule) | /n | ?4 ? PROCEED leg;
?3 ? PIVOT A leg |
154 | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
155 | H3 GREEN-or-AMBER (audience exists) | /n | ?6 ? PIVOT A leg |
156 | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
157 | H1 any recording | /n | <3 ? FALSIFIED leg |
158 | H5 RED **and** Q3 names parents/players | /n | ?4 ? PIVOT C |
159
160 ## 5. Guardrails that bind the field work
161
162 - **Phase 1 records nothing** ? notes only (playbook rule 7). Student
163 conversations: coach-permitted **adult (18+) students only** ? "senior batch"
164 minors don't qualify; skip and note it when no adults are available.
165 - **Phase 2 consent rules are hard:** written consent from the academy AND adult
166 players; **sessions with minors are never recorded for our library** (training-pool
167 guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
168 - Answer sheets contain personal contact details ? they stay between the associate
169 and the owner; transcribe to the rollup without phone numbers.
170

```

17. assistant (2026-06-21T13:56:44.352Z)

Now let me read ROADMAP and BACKLOG, and check NEXT_STEPS for the prioritized pick-up.

18. user (2026-06-21T13:56:45.304Z)

```

1 # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3 This document is the **narrative thread**: how the indexer evolves from an
4 AI-API-dependent prototype into a self-contained tool that learns rally detection
5 from its own commercially-licensed, annotated video corpus. It explains *how we
6 got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
7 are recorded as ADRs in [DECISIONS.md](archives/decisions/DECISIONS.md); the scoring
mechanics live
8 in [EVALUATION.md](EVALUATION.md).
9
10 > ? **Phase 4 update:** the authoritative owned-model path is now
11 > [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? a learned
**per-frame TAL head**
12 > (ActionFormer/ASFormer over WASB trajectory features), not the candidate-window
classifier sketched below.
13 > **Audio is parked, not "gated on golden labels":** the E1 probe and a follow-up
foreground-audio test both came back
14 > weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's
Gen-1 note).
15
16 ## The thesis
17
18 The indexer detects badminton rallies in long recordings. Its strongest mode

```

```

19  (`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
20  player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
21  tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
22  works, but it costs money, needs the network, and never improves on its own.
23
24  > *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
25  > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
26  > ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
27  > player-motion gating stage, not the Ultralytics library.
28
29  A sibling project, **sports-data-collector**, curates sports videos with
30  per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
31  this roadmap: **use that annotated, commercially-usable data to (1) measure the
32  pipeline objectively and (2) train an offline replacement for the paid Gemini
33  stage** ? turning the tool into one that improves as more data is curated, runs
34  air-gapped, and generalizes across sports.
35
36  Why it matters: it removes the per-run API cost and network dependency, gives us
37  a defensible quality metric instead of eyeballing, and creates a flywheel ?
38  more curated matches ? a better offline segmenter ? better highlights.
39
40  ## The four phases
41
42  ### Phase 1 ? Bridge the repos (DONE)
43  A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
44  ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
45  the `is_commercial` flag. ? *Why:* one clean producer/consumer contract that works
46  for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.
47  See **ADR-002**.
48
49  ### Phase 2 ? Acquire full-resolution video (DONE)
50  The corpus was 256x144 (the original crawl used `worstvideo`). The `fetch` command
51  re-acquires every video at **best available resolution (1080p here)**, **in place**
52  (preserving annotations), via **authenticated Firefox Premium cookies** and yt-dlp's
53  `default` client. ? *Why:* YOLO/shuttle detection needs real pixels; and the
54  journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate
55  limiting) now encoded in the tooling. **22/22 fetchable matches are now 1080p**.
56  See **ADR-003**.
57
58  ### Phase 3 ? Evaluate + tune (IN PROGRESS)
59  `scripts/run_phase3_eval.py` walks the manifest, processes each video, and scores
60  detected
61  rallies against the ground truth ? per video and corpus-aggregate (precision/recall/
62  F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final`
63  (refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error
64  lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation**
65  problem (candidates catch them, refinement is loose). That diagnosis decides where
66  effort goes, and establishes the baseline any trained model must beat.
67
68  ### Phase 4 ? Learn an offline, API-free segmenter (NEXT)
69  Replace the paid Gemini refinement with a **lightweight classifier trained on the
70  rally annotations**, consuming features the pipeline already computes and stores
71  (`candidate_segments.stats`). No external API, no new labeling. ? *Why:* this is the
72  payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from
73  data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate
74  stage at **recall ~0.05 @ IoU 0.5** on `shuttleaset_8` (median best coverage of a
75  rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's
76  substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.
77
78  ## Complementary local cues (beyond the segmenter swap)
79
80  Once the substrate is WASB shuttle-motion, the next lever is **fusing additional
81  fully-local cues** to raise rally recall/precision without reintroducing a paid
82  API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**,
  accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in

```

```

83     `backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ?
84     verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover
85     WASB-missed rallies) but the classical-onset precision ceiling (~2.2x) is too weak
86     to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a
87     *learned timbre classifier* ? but per the owned-model plan audio is now **parked** (the
88     foreground-audio idea also failed), **not** revived now that labels exist.
89     See `archives/research/AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is the more
credible #2 cue.
90
91     The recurring blocker across Phases 3?4 and the audio work is the same: **more
92     golden labels** is the single highest-leverage move (see the cross-cut note below).
93
94     ## How this shapes the final tool
95
96     - **Offline & free by default.** The production rally mode becomes a local model;
97     Gemini is kept only as an upper-bound reference, not a dependency.
98     - **Self-improving.** Every newly curated, annotated match is more training signal;
99     re-training is a local, deterministic step.
100    - **Measurable.** Every change is scored against the golden set, so "better" is a
101    number, not a vibe.
102    - **Sport-generic.** The label ? feature ? classifier machinery and the export
103    bridge already span sports; adding a sport is data, not new architecture.
104    - **Commercially clean.** Only `is_commercial` data flows through the bridge, so the
105    trained model and any derived product rest on a licensable foundation.
106
107    ## Status at a glance
108
109    | Phase | What | Status |
110    |-----|-----|-----|
111    | 1 | Export bridge (`curate export` ? manifest + CSVs) | ? Done |
112    | 2 | 1080p acquisition (`fetch`, in-place, cookies) | ? Done (22/22) |
113    | 3 | Eval + tuning loop (`scripts/run_phase3_eval.py`) | ? In progress |
114    | 4 | Offline learned segmenter (WASB shuttle-motion substrate) | ? Next (substrate
decided: WASB, ADR-004) |
115    | + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1 probe run ?
AMBER (~0.6 AUC), parked / de-prioritized |
116
117    > **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand
regression testing)
118    > is produced via a vendor process + pluggable annotation provider ? see
119    > [GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md) and
**ADR-006**. As of 2026-06-02 the annotation
120    > guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is
selected for an
121    > initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality
requirements
122    > that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](data_pipeline/VIDEO_RECORDING_GUIDELINES.md).
123    >
124    > **Observability:** every processed video now emits an always-on per-video diagnostics
125    > report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
126    > pipeline regressions and footguns surface without re-running eval ? see
127    > [CODE_MAP.md](CODE_MAP.md) §2 (reporting/observability).
128

```

19. user (2026-06-21T13:56:45.406Z)

```

1     # Backlog ? deferred items to revisit
2
3     Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5     > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to **GitHub issues** instead (or in addition).

```

```

6      > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8      _Last updated: 2026-06-10._
9
10     ---
11
12     ## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
13     `NEXT_STEPS.md`)
14     - ? **Rename the repo off "badminton"*** ? owner-emphatic "ASAP" action; the project is
15     multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit:**
16     owner action, before broad external sharing.
17     - ? **Quantify cost-to-Gen-2 (the largest open risk)** ? benchmark real VRAM/token/$ for
18     the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 678 GPUs)
19     **before** standing up any owned serving stack. **Revisit:** before Phase-2 commitment.
20     - ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized
21     (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4
22     harness.
23
24     ## Engineering hardening deferred from the 2026-06-10 audit sweep
25     - ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app`
26     leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
27     (#16). **Revisit:** when the async slice starts.
28     - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
29     `*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
30     **Revisit:** before any volume/batch processing.
31     - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):**
32     `detectors/base.py:~wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
33     tilde expansion coexist; applied to every `~`-capable arg in both runners;
34     `tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
35     `~wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
36     default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
37     (repo_dir/conda_env) if config ever becomes per-tenant.
38
39     ---
40
41     ## Monetization / licensing
42     *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
43
44     - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
45     torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not
46     needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
47     YOLOv8.
48     - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
49     Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
50     "small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
51     output. **Revisit:** before commercial launch.
52     - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
53     ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms.
54     Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
55     weights.
56     - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
57     ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
58     the product. **Revisit:** when choosing the production LLM step.
59
60     ## Detector swap follow-ups
61     *(source: PR #8, ADR-005)*
62
63     - ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
64     supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
65     stopgap. Migrate to `ByteTracker` from the separate `trackers` package before lifting the
66     cap. **Revisit:** before bumping supervision past 0.30.
67     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
68     tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
69     (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
70     expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.

```

```

37     - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
38
39     ## Golden-set subproject
40     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
41
42     - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
43     - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
44     - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
45     - ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
**Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).
46     - ? **GS-5 ? regression triggers** ? wire `regress()` into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
**Revisit:** after GS-4.
47     - ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor
pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as
calibration once the gold key exists.
48     - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
**? held out by match**. **Unblocked.**
49     - ? **Train/eval separation guardrails** (was `DEFERRED_HARDENING_PLAN` #13; design
preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs**
today, kept apart by convention only; nothing *enforces* that a video used to train the learned
segmenter (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design**
(from the archived DEFERRED plan §2): split unit = **match, never clip** (hold ~? of each sport
as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md`); add a `split: "train"|"eval"` field to the
collector's `curate export` manifest (cross-repo contract, ADR-002); **[DONE ? pure-logic core,
PR #171]** `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids,
eval_ids)` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in
`compute_video_id`) + `assert_disjoint(train_ids, eval_ids)` that raises on overlap, with
`tests/test_splits.py` (overlapping ids ? `assert_disjoint` raises). **STILL DEFERRED:** wire
the guard into the training entrypoint (`distill_local`/`training`) and into `run_regression.py`
(baseline computed strictly on the eval split); add the collector's `curate export` manifest
`split` field (above); the integration test (a deliberately-leaked manifest fails the training
entrypoint). Full rationale/risks: [archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2.
**Revisit:** dedicated design session with the user (needs a collector-side manifest change).
50
51     ## Recording best practices
52     *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
53
54     - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.
55
56     ## Code & test organization
57     *(source: owner review comments on PR #35, 2026-06-07)*
58
59     - ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test

```

components on a grouping

60 metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
`backend/pipeline/audio/` +

61 `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
tests under `tests/eval/`).

62 The exact grouping ****metric is the owner's design choice**** (consolidate small files
vs. add per-area dirs ? the

63 "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
****dedicated refactor PR****,

64 not guessed here. ****Revisit:**** when the owner specifies the grouping scheme. (The
`backend/eval/audio/` half of

65 the request ? comment 1 ? was done in PR #35.)

66

67 ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)

68

69 *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
second-in-command)*

70

71 - ? ****Docs-only checkpoint PR**** ? This update + the refreshed
`docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (since archived)
formally record the approved path before any code changes. The initial review docs (PR #37)
introduced the findings; this entry locks the execution decisions.

72 - ? ****Approved directory structure**** ? Adopt a layered approach under `backend/` for
professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-
file dirs):

73 - `backend/config/` (package ? start with this)

74 - `backend/api/` (FastAPI app, routes, static UI)

75 - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)

76 - `backend/infrastructure/` (database + external adapters)

77 - Keep `backend/eval/` prominent as a first-class concern.

78 - Logical grouping for providers, sports, annotations, utils, and a small `core/` only
when needed.

79 - ? ****Sequencing (Config first, then atomic refactor)**** ? Implement Deliverable 1
(Config) ****first****, introducing the `backend/config/` package immediately. Perform the full
directory structure refactor as a ****separate high-pri atomic PR**** afterwards. This minimizes
conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.

80 - ? ****Deliverable 1 (Config)**** ? `backend/config/` package (`models.py` + `loader.py`)
instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file
override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this
before the layout move.

81 - ? ****Subsequent items**** ? See the refreshed `docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging,
packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its
own high-pri step right after Config.

82

83 ****Status legend reminder:**** ? = approved / in progress tracking, ? = do when triggered.

84

85 ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)

86 *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
#11/#12; shared lib `obsreport`)*

87

88 - ? ****Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are
DONE.**** Remaining = items 3?9 in `docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived).

89 - ? ****Per-video observability reports shipped in BOTH tools**** (a TECHNICAL report + a
CUSTOMER-friendly summary), via the ****soft-dependency**** lib `obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `.report.md / .summary.md` next to the video.

90 - ? ****Item 8 (Input-Quality UX) is LARGELY SUBSUMED**** by these reports (validation stage
+ footgun flags + verdict already surface quality; a clean customer summary already exists).
****Do NOT build a parallel mechanism**** ? instead EXTEND the report: add an audio-readiness
stage/flag to `backend/reporting/` (harvest) + the obsreport spec. ****Revisit:**** when item 8 is
picked up.

91 - ? ****Item 7 (unify hybrid segmenters) should land before the report's real AI-provider
timing.**** The report records AI timing as `not\_captured`; once `wasb\_hybrid/` tracknet_hybrid`

are unified there is ONE place to thread a recorder callback through `provider.analyze\_video` ? the report. ****Revisit:**** with item 7 (Claude to add the timing hook there).

92 - ? ****Item 3 (Error/Result contract) synergizes with the report.**** Today the report INFERS failures (e.g. `silent\_provider\_noop` = 0 segments + no API key). A structured Result/error from providers/segmenters would let the report read the real reason as a first-class flag. ****Revisit:**** with item 3.

93 - ? ****`obsreport` is a SOFT/commented dep**** in `requirements.txt` (works locally via editable install; pipeline no-ops if absent). ****Pin the `git+ssh@v0.1.3` tag**** in both repos once SSH/CI access to the private lib repo exists.

94 - ? ****`run\_all\_tests.py`**** added at repo root ? run the full suite (clear PASS/FAIL, pytest exit code) before sending PRs.

95 - ? ****Config-migration follow-ups (DONE ? config & identity consistency pass)**** ? the two typed-config bugs from

96 #39/#40 are fixed: (1) ****`/api/compile` now honors `--output-dir`**** and (2) ****`output\_dir` is a real field on**

97 ****`OutputConfig`**** (default "output"); the CLI overrides it on the typed model in ****`main()`****, retiring the

98 write-only ****`\_runtime\_overrides`**** hack. Same pass also unified ****`video\_id` MD5 derivation into**

99 ****`backend/utils/hashing.py::compute\_video\_id`**** and migrated all four ****`run\_\*.py`**** scripts off raw ****`json.load`**** onto

100 ****`backend.config.load\_config`****. See ****`CODE\_MAP.md` S2.0/S6 (footguns retired).**

101

20. assistant (2026-06-21T13:56:56.875Z)

Now let me check the live NEXT_STEPS doc since it's referenced as the prioritized pick-up and supersedes parts of these. Let me also check DOC_STATUS for any status updates on these docs.

21. user (2026-06-21T13:56:57.295Z)

1 # Next steps (live) ? refreshed 2026-06-21

2

3 > ****How to read:**** this top section is the ****current cross-track state + prioritized pending list**** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed ****owned-model & rally-quality track**** plan (2026-06-13) is ****preserved below, still valid**** ? it is now ****one of several* active tracks**. Live blow-by-blow: memory ****`current-state.md`**. Doc lifecycle/hygiene: ****`docs/DOC\_STATUS.md`**.

4

5 ****`## ? Where we are (2026-06-21)`**

6 ****`DECOUPLED\_COMPUTE` delivered on GCP + archived**** (#270); ****`compute` = GCP only**** (ADR-013, DO dropped). ****`COMPUTE\_DECOUPLED\_SERVING`**** subproject spun up (#272, ****`ADR-014`****) ? one pure core + local/cloud profiles; ****`owner` M0 design sign-off pending****. ****`Per-video workspace P0`**** shipped (#264); ****`doc-status` register + archival due-diligence**** in place (#273). The ****`khelsutra` site + per-rally UI**** ship in parallel (sibling track). The ****`rally-quality / owned-model`**** track (below) is the core ML work, gated on ****`growing the golden corpus`****.

7

8 ****`## ? Prioritized pending ? all tracks`**

9 ****`P0 ? highest leverage / unblocks the most`****

10 1. ? ****`COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED (#279/#280/#282/#283) + M5+ batch authorized**** (owner 2026-06-21, ADR-014 ratified). All \$8 gating Qs locked (D7?D17). ****`Owner residual`**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****`\$100/?10k cap`**** (D17).

11 2. ****`[owner, ongoing] Grow the golden corpus`**** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ****`and`**** full-corpus training feed on it.

12 3. ****`[owner + Claude + GPU] Full-corpus training (n?5)`**** ? a real ****`weights/<ver>`**** (the ****`0.1.0-l4`**** bundle is n=2 plumbing-only).

13

14 ****`P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)`****

15 4. ****`[Claude] COMPUTE\_DECOUPLED\_SERVING` M5?M12 default-OFF code batch`\*\*** (owner-authorized 2026-06-21, merge-each-on-green): M5 truly-local preset+startup-checks+L4 parity test ? M6 **\*\*`backend/compute/`** Cloud Run dispatch + Dockerfile (mocked) ? M8 v6 ledger+pricebook+/cost ? M9-auth Cf-Access JWT+tenant-scoping ? M11 flywheel plumbing ? M12 **\*\*`gdrive-api`**** backend.

```

*(M1?M4 config/input/output/device layer = ? SHIPPED.)*
16      5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video
routing of `detector/`+`tmp/` *[the `output/chunks_` de-hardcode base = ? done via #282]* ? P4
? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-
ledger migration ? author the version bump once.
17      6. **[Claude] Ops-Agent codification into `deploy/gcp`** (every VM auto-enables
observability ? [[gcp-ops-agent-enable]]).
18      7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
19      8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the
ablation) ? needs GPU-extracted golden features.
20
21      **P2 ? later / gated / deferred**
22      9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
23      10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
24      11. **Multisport** (table-tennis first, then pickleball).
25      12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
26      13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
27      14. **Rename repo off "badminton" (owner, BACKLOG).
28      15. Owner housekeeping: retire the shared **D0 droplet** + **D0 token**; collector
**`md5`** export (collector repo).
29      16. **PMF field validation** (owner-side, business track).
30
31      **Doc-hygiene (tracked in `docs/DOC_STATUS.md` $4):** ? #1 per-video reconcile + ? #2
this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.
32
33      ---
34
35      ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
36
37      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
38      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
39      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
40      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;  

**PR-2
41      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
42      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
43      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
44      `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
45      **Authoritative roadmap:**
46      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
47      `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
48
49      ## ? Next-session / GPU-box pickup (2026-06-15)
50      **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
51
52      **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden

```

```

features + `.rallies.csv` labels are not yet here):
53     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
54     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
55     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ? F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
56
57     **Also queued (owner-gated / pending input):**
58     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
59     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
60     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
61
62     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
63
64     ## Where we are (one paragraph)
65     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
66     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
67     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
68     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
69     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
70     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
71     battery is done** (see table) ? the moat is quantified.
72
73     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
74     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
75     |---|---|---|
76     | Heuristic (velocity windowing) | 0.657 | 0.157 |
77     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
78     | Two-stage WASB+Gemini refine | 0.83 | ? |
79     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
80
81     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
82     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
83     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
84     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
85     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
86     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
87
88     ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
89     **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.

```

90

91 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `\docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `\docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

92

93 - **Docs:** `\docs\MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) .

94 `\docs\EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

95 `\docs\ROORA_LABELING_GUIDELINES.md` (v8) . `\docs\HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).

96 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `\backend\pipeline\detectors\rally_gate.py` already implements the net-crossing `**"Gate A"*` (`\apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `\min_crossings` knob in `\IndexingConfig`). **Wiring Gate A into the live segmenter (now via `\fusion_hybrid`) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `\fusion_hybrid`, `**default-OFF**`, seeds from substrate, persists `\net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `\tests\test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

97 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `\PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `\compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via `\hit_detector`.

98 - **Pick-up order (each = ONE PR off `\master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**

99 1. (P2 first-step, largely done via `\fusion_hybrid`) Wire Gate A into production served path + config knob (`\min_crossings`); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

100 2. P3 first step (this work item): harness `\compare` LOVO litmus for person-on variant (`\players_in_court` + `\two_sided` + `\colocation` etc. from `\fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

101 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

102 - **Discipline (from `\EXPERIMENT_HARNESS.md`):** every new cue `**flag-gated, default OFF**`; promote to default `**only on measured LOVO lift**` through `\run_regression.py` (exit 0/1/3); pinned `\CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

103 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `\.rallies.csv` labels, which are not yet present in this checkout.

104 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in `\RALLY_QUALITY_RESEARCH.md` (`\RALLY_QUALITY_RESEARCH.md`) §5.6.** An owner-commissioned 31-reference review of the quality report `*validated*` the methodology and surfaced these `**future**` levers (research inputs, not active work, not new claims; verify citations before any submission):

105 1. `**"First-mile" paper framing (highest-value):**` stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all `**assume a pre-trimmed rally clip already exists**` ? we solve the ignored prerequisite. Pitch the paper as `**the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage.**`

106 2. `**Ground R2 in Action-Spotting / Precise-Event-Spotting**` (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

107 3. `**Add domain baselines for any submission:**` MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

108 4. `**Prefer TemporalMaxer over ActionFormer**` for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

109 5. `**Court polygons ? pose ? the rally-END lever:**` masking on-court players can resolve `*occluded end-of-rally states*` (the #1 remaining gap vs Gemini), not just spectator noise.

110 6. `**Name the eval?serve lesson "pipeline distribution skew"***` ? a citable paper "lesson."

111

```

112     **Discipline reminder:** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
113
114     ## Prioritized next steps
115     1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
116         Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
117         lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
118         corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
119         `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
120         label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
121         The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
122     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
123         reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
124         (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
125         paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
126     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal
head (MIT, minutes on the
127         2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the
128         LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer
TemporalMaxer** (parameter-free
129         max-pool, ~2.8x fewer GMACS, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review
130         (`RALLY_QUALITY_RESEARCH.md` $5.6) ? better fit for the cheap/local/fast mandate.
131     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
132         badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
133         by C3 (provenance multiplies per sport).
134     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
135         "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
136         (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
137         `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
138         physical-visit approach; reframed product-feedback-first per owner spec):**
139         `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
140         `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
141         rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
142         capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
143         decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
144     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
145         deletion in AI Studio); **rename repo off "badminton"*** (owner action,
146         in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
147
148     ## Pointers / artifacts

```

```

149      - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
150      Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
151      `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
152      - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
153      manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
154      - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
155      (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
156      labels: `...\Collect\Golden Labelled\`.
157      - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
158      (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
159      with a MISLEADING "credits depleted" 429; ramp gradually.
160      - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
161      PLATFORM slice = async job model (single-tenant).
162

```

22. assistant (2026-06-21T13:57:02.497Z)

I now have a comprehensive picture. The NEXT_STEPS doc is the live prioritized pick-up and reveals significant updates beyond what the static plan docs show (e.g., M0 in-container quick wins #170/#171/#174 are done, Gen-0 harness ships with LOVO 0.626, the Gemini battery is done). Let me verify a couple of the status claims against the actual code/artifacts to flag any staleness ? specifically the Gemini-silver purge (M0.3) and whether the Gen-0 harness exists.

23. user (2026-06-21T13:57:05.677Z)

```

=== distill_local.py (M0.3 purge target) ===
-rw-r--r-- 1 avidu 197609 8602 Jun 14 16:36 backend/eval/distill_local.py

=== Gen-0 harness ===
-rw-r--r-- 1 avidu 197609 12222 Jun 21 08:12 training/gen0/harness.py

=== splits.py (train/eval guard, #171) ===
-rw-r--r-- 1 avidu 197609 3647 Jun 15 14:11 backend/eval/splits.py

=== eval_baselines ===
experiment_leaderboard.json
fixtures
gemini_wrapper_2026-06-10.json
heuristic_lovo_n6.json
nightly_baseline.json

```

24. assistant (2026-06-21T13:57:09.537Z)

The Gen-0 harness exists (most recently modified, Jun 21), splits.py exists (#171 train/eval guard), and the baselines are recorded. Note `distill\_local.py` still exists (M0.3 purge not yet executed ? it's gated on "before any training run"). Let me quickly confirm whether the M0.1 event-index DB migration has happened (the conversational seam), since that's a key gated item.

25. user (2026-06-21T13:57:10.822Z)

```

69:      ``user_version`` stays un-bumped ? and the next open would re-run the same
88:      # Versioned migrations live here. PRAGMA user_version is the schema
91:      version = cursor.execute("PRAGMA user_version").fetchone()[0]
110:      # future: if version < 7: cursor.execute("ALTER TABLE ..."); PRAGMA user_version
= 7
183:      cursor.execute("PRAGMA user_version = 1")
210:      cursor.execute("PRAGMA user_version = 2")
222:      cursor.execute("PRAGMA user_version = 3")

```

```
242:         cursor.execute("PRAGMA user_version = 4")
280:         CREATE UNIQUE INDEX IF NOT EXISTS idx_user_models_user_version
290:         cursor.execute("PRAGMA user_version = 5")
337:         cursor.execute("PRAGMA user_version = 6")
```

26. assistant (2026-06-21T13:57:14.674Z)

The DB is at user_version 6, but there's no `highlights` table, `fault\_tags`, `rally\_ordinal`, or `ending\_reason` column yet ? confirming the M0.1 event-index migration (the conversational seam) has NOT been built (consistent with NEXT_STEPS line 62 listing "event-index DB migration (M0.1)" under "Leaving (bulky / M2 ? no start without owner go)"). I now have everything needed to produce the structured output.

27. user (2026-06-21T13:58:22.665Z)

Structured output provided successfully

28. assistant (2026-06-21T13:58:34.291Z)

I've completed the analysis of the owned-model + strategy track and returned the structured output. Summary of what I found:

****Status:**** Owned-model track is at Phase 0, partially executed. The Gen-0 TAL harness ships (verified `training/gen0/harness.py`, modified today) and produced artifact v0.1.0 at LOVO F1 ****0.626**** ? passed the 0.480 heuristic floor but the sign test (5/6 wins, p=0.109) is not significant, so `ship=False`. The n=6 corpus is grown, the Gemini battery is complete (multi-court wrapper 0.66 = the owned model's ship target; clean footage 0.93 conceded to Gemini), and the train/eval split guard core landed (#171).

****Still gated / not started**** (verified against code): M0.1 event-index DB migration is NOT done (DB at `user\_version=6`, no `highlights` table / `rally\_ordinal` / `fault\_tags` / `ending\_reason`); M0.3 Gemini-silver purge pending (`backend/eval/distill\_local.py` still present); M0.4 TemporalMaxer/ActionFormer scorer swap; real P3 person-fusion LOVO (blocked on GPU-extracted golden features).

****DEFERRED_HARDENING_PLAN is fully completed**** ? #16 async jobs (PR #87), #7 packaging (PR #167), #13 train/eval split moved to BACKLOG.

****Owner gates**** captured: Phase-0 milestone greenlight, training-data opt-in/minors-exclusion, ship-gate target (#172), cost-to-Gen-2 budget, WASB C3 + codec legal sign-off, PMF field validation, repo rename.

****Staleness flagged**** (not fixed): ROADMAP Phase-4 narrative superseded by the TAL plan; COMMERCIALIZATION still calls async/packaging "pending approval" though both are done + has link rot to archived PMF/vendor docs; the authoritative plan's M0.4 still says ActionFormer while NEXT_STEPS prefers TemporalMaxer; BACKLOG header predates the Gen-0 artifact.